Crash Job Detection & Recovery Timing Report

TCS Traffic Control Service • Log-derived measurements • 17 October 2026

Observed log window: 2026-10-17T20:39:45.182+05:30 to 2026-10-17T20:43:28.154+05:30. Duration: 222.972 seconds. Source file supplied in this conversation; no outside assumptions were used.

Executive summary

Observed item	Count	Evidence / interpretation
Distinct jobs with admission + crash-recovery trace	15	All observed job IDs appear in admission, worker, recovery, and final-discard logs.
Worker pickups / attempts	30	Two pickups are logged for each job: initial attempt and retry.
Simulated exception markers	30	The log text identifies the injected exception as “Simulated Exception during heart beat”.
Crash recovery script marker pairs	30	Each attempt has a recoverOneJob start marker and a recovery-result marker.
Jobs marked for retry / max-retry discard	15 / 15	Classification wording is taken directly from CrashJobRecoveryService logs.
DLQ entry logs / Hibernate failed_jobs INSERTs	15 / 15	SQL statements have no per-statement timestamps, so they are not assigned an execution duration.
Final “Job discarded” markers	15	All final discard markers include an explicit epoch-millisecond timestamp.

Measured timing results

Interval	Observed distribution	Caveat
Requested sleep-start → sleep-end marker, all attempts	median 1.200 s; range 1.200 s–1.201 s (n=30)	25 readings at 1,200 ms and 5 readings at 1,201 ms across all 30 attempts.
Exception marker → heartbeat-stop call, both attempts	median 250.0 ms; range 250 ms–251 ms (n=30)	250 ms typical; range reflects millisecond timestamp values in the logs.
Initial heartbeat-stop marker → first recovery attempt	median 6.182 s; range 6.059 s–6.233 s (n=15)	Based on embedded Java epoch milliseconds at both endpoints.
Initial exception marker → first recovery attempt	median 6.432 s; range 6.309 s–6.483 s (n=15)	Elapsed time until the scheduler’s first recovery attempt; not the worker exception-handling duration.
Recovery-result log → retry pickup log	median 11 ms; range 0 ms–53 ms (n=15)	Uses logger timestamps at both endpoints; pickup and recovery occur on different threads.
Retry heartbeat-stop marker → final recovery attempt	median 3.540 s; range 3.496 s–3.550 s (n=15)	Based on embedded Java epoch milliseconds at both endpoints.
recoverOneJob start → result marker, first pass	median 2 ms; range 0 ms–3 ms (n=15)	Marker-to-marker interval with millisecond resolution; not a nanosecond benchmark.
recoverOneJob start → result marker, second pass	median 2 ms; range 0 ms–3 ms (n=15)	Marker-to-marker interval with millisecond resolution; not a nanosecond benchmark.
DLQ-entry log → final discard log	median 8 ms; range 6 ms–10 ms (n=15)	Includes everything between those application log markers; not pure MySQL execution time.
Initial query-start epoch → final discarded epoch	median 12.646 s; range 12.524 s–12.696 s (n=15)	Full observed worker/recovery/discard path from the embedded query-start marker to the embedded final-discard marker.

Interval	Observed distribution	Caveat
Admission-attempt logger timestamp → final discard logger timestamp	median 12.659 s; range 12.577 s–12.775 s (n=15)	End-to-end logger-to-logger interval, including admission and worker/recovery processing.

Timestamp and measurement rules

- For intervals whose two endpoints contain explicit Java epoch-millisecond values, calculations use those raw integers; each is converted by adding that many milliseconds to the UTC epoch and then displaying it in Asia/Kolkata (UTC+05:30).
- The ISO logger timestamp and the embedded Java timestamp are kept as separate fields. For example, line 3 logs at 20:39:45.262 IST while its query-start value 1792249785261 converts to 20:39:45.261 IST; the 1 ms difference is preserved, not “corrected.”
- The “sleep ends at” log line does not include a job ID. For the detailed CSV only, it is associated with the immediately following “Exception throw called for jobId” line when both lines name the same worker thread. The association rule is written in the CSV.
- The ERROR log line also omits a job ID; it is associated only with the immediately preceding exception-marker line on the same worker thread.
- Two retry query-start epoch values are 2 ms earlier than their preceding recovery-result epoch markers, while their worker pickup log records appear after the recovery-result log records. These are shown as raw marker deltas in the CSV and are not described as negative elapsed log-to-log time. Use the logger-to-logger measurement for the retry-pickup interval.
- Hibernate SELECT/INSERT statements have no timestamps. The report gives their source line locations and reports only the measured time between surrounding application log markers; it does not claim a MySQL execution or transaction-commit duration.

Per-job sequence: first detection and retry

Times in this table are same-day clock times (17 October 2026, IST). The first pickup uses the embedded query-start epoch converted to IST; the retry pickup is the logger timestamp of the worker “Job found” line. Full IDs and raw epoch values are in the CSV.

Job ID	Initial query-start (IST)	Attempt 1 heartbeat-stop (IST)	Recovery result 1 (IST)	Retry pickup log (IST)	Stop → recovery attempt	Result log → retry log
01a14a69-6b5e-72f3-9656-34bf7c4ef62a	20:39:45.261	20:39:46.714	20:39:52.947	20:39:53.000	6.230 s	53 ms
01a14a69-a631-7b61-b28f-c21b3b741a76	20:40:00.276	20:40:01.729	20:40:07.961	20:40:07.972	6.229 s	11 ms
01a14a69-e0e8-7072-9995-75f6e0cf5b06	20:40:15.288	20:40:16.740	20:40:22.974	20:40:22.979	6.233 s	5 ms
01a14a6a-1b98-76c0-9607-51c2104b5e7e	20:40:30.303	20:40:31.755	20:40:37.987	20:40:38.040	6.230 s	53 ms
01a14a6a-564b-7d98-8f13-2da7735f1c11	20:40:45.419	20:40:46.872	20:40:53.002	20:40:53.002	6.128 s	0 ms
01a14a6a-9102-7cec-9d57-99a8399d918f	20:41:00.381	20:41:01.832	20:41:08.016	20:41:08.026	6.182 s	10 ms
01a14a6a-cbae-71a9-9d9f-f065e0934699	20:41:15.388	20:41:16.840	20:41:23.030	20:41:23.034	6.188 s	4 ms
01a14a6b-065a-70dc-805a-dc53d4161096	20:41:30.402	20:41:31.853	20:41:38.041	20:41:38.044	6.188 s	3 ms
01a14a6b-4107-721b-aeef-172d86a5810c	20:41:45.415	20:41:46.866	20:41:53.061	20:41:53.095	6.192 s	34 ms
01a14a6b-7bbc-7e1d-a789-fcd11733cb98	20:42:00.456	20:42:01.908	20:42:08.072	20:42:08.108	6.164 s	36 ms
01a14a6b-b677-7adb-82a9-cd8b06fbce73	20:42:15.482	20:42:16.932	20:42:23.086	20:42:23.093	6.151 s	7 ms
01a14a6b-f12a-7598-b8fe-944c74300e5c	20:42:30.572	20:42:32.023	20:42:38.099	20:42:38.129	6.075 s	30 ms
01a14a6c-2bd9-7344-9cb0-abd3284f1276	20:42:45.564	20:42:47.015	20:42:53.114	20:42:53.114	6.096 s	0 ms
01a14a6c-6685-7f8b-87b9-0b1517c3c535	20:43:00.571	20:43:02.023	20:43:08.128	20:43:08.149	6.104 s	21 ms
01a14a6c-a139-7fbc-8973-3ebfc7632c35	20:43:15.630	20:43:17.081	20:43:23.143	20:43:23.176	6.059 s	33 ms

Measurement note: the “heartbeat-stop → recovery attempt” interval uses explicit epoch values from the stop-call and scheduler-attempt messages. “Result log → retry log” uses the two logger timestamps so the order observed in the log output remains non-negative.

Per-job sequence: retry exhaustion and DLQ

Final recovery and discard timestamps are taken from the explicit embedded epoch values where present; the DLQ-entry time uses its ISO logger timestamp because that log line has no embedded epoch value.

Job ID	Attempt 2 heartbeat-stop (IST)	Recovery result 2 (IST)	Max-retries classification (IST)	DLQ entry log (IST)	Final discarded (IST)	Stop → final recovery	Initial pickup → discard
01a14a69-6b5e-72f3-9656-34bf7c4ef62a	20:39:54.451	20:39:57.950	20:39:57.950	20:39:57.950	20:39:57.957	3.496 s	12.696 s
01a14a69-a631-7b61-b28f-c21b3b741a76	20:40:09.422	20:40:12.964	20:40:12.964	20:40:12.964	20:40:12.972	3.540 s	12.696 s
01a14a69-e0e8-7072-9995-75f6e0cf5b06	20:40:24.430	20:40:27.977	20:40:27.977	20:40:27.977	20:40:27.984	3.544 s	12.696 s
01a14a6a-1b98-76c0-9607-51c2104b5e7e	20:40:39.491	20:40:42.991	20:40:42.991	20:40:42.991	20:40:42.998	3.497 s	12.695 s
01a14a6a-564b-7d98-8f13-2da7735f1c11	20:40:54.453	20:40:58.003	20:40:58.003	20:40:58.004	20:40:58.012	3.550 s	12.593 s
01a14a6a-9102-7cec-9d57-99a8399d918f	20:41:09.476	20:41:13.019	20:41:13.020	20:41:13.020	20:41:13.027	3.541 s	12.646 s
01a14a6a-cbae-71a9-9d9f-f065e0934699	20:41:24.485	20:41:28.033	20:41:28.033	20:41:28.034	20:41:28.040	3.546 s	12.652 s
01a14a6b-065a-70dc-805a-dc53d4161096	20:41:39.494	20:41:43.045	20:41:43.045	20:41:43.045	20:41:43.053	3.548 s	12.651 s
01a14a6b-4107-721b-aeef-172d86a5810c	20:41:54.545	20:41:58.064	20:41:58.064	20:41:58.064	20:41:58.071	3.516 s	12.656 s
01a14a6b-7bbc-7e1d-a789-fcd11733cb98	20:42:09.559	20:42:13.075	20:42:13.075	20:42:13.076	20:42:13.082	3.514 s	12.626 s
01a14a6b-b677-7adb-82a9-cd8b06fbce73	20:42:24.544	20:42:28.089	20:42:28.089	20:42:28.089	20:42:28.097	3.542 s	12.615 s
01a14a6b-f12a-7598-b8fe-944c74300e5c	20:42:39.580	20:42:43.102	20:42:43.102	20:42:43.102	20:42:43.110	3.519 s	12.538 s
01a14a6c-2bd9-7344-9cb0-abd3284f1276	20:42:54.565	20:42:58.117	20:42:58.117	20:42:58.117	20:42:58.126	3.550 s	12.562 s
01a14a6c-6685-7f8b-87b9-0b1517c3c535	20:43:09.599	20:43:13.129	20:43:13.129	20:43:13.129	20:43:13.139	3.529 s	12.568 s
01a14a6c-a139-7fbc-8973-3ebfc7632c35	20:43:24.626	20:43:28.146	20:43:28.146	20:43:28.146	20:43:28.154	3.518 s	12.524 s

Direct observations from the logs

- Every one of the 15 job IDs has two worker pickup records, two injected-exception markers, two heartbeat-stop calls, and two recovery-result markers.
- The first recovery classification says the job “has been identified as a crash job and will be retried” for 15 jobs. A later classification says “Max retries exhausted and job is getting discarded” for 15 jobs.
- The log contains 15 “Adding entry to DLQ” records and 15 untimestamped Hibernate INSERT statements targeting failed_jobs.
- The final discard log reports ttl:600 for each job. This is reported as the literal log value; the report does not infer TTL expiry behavior beyond the line itself.
- Observed final-path duration from initial query-start marker to final discarded epoch has a median of 12.646 seconds and a range of 12.524–12.696 seconds.

Source traceability

The report was generated from the supplied text log (795 logical lines). First timestamp: 2026-10-17T20:39:45.182+05:30. Last timestamp: 2026-10-17T20:43:28.154+05:30. The detailed CSV lists source line numbers for the event markers and, separately, both logger timestamps and embedded Java epoch values. The final discard SQL area has no per-statement timing information.